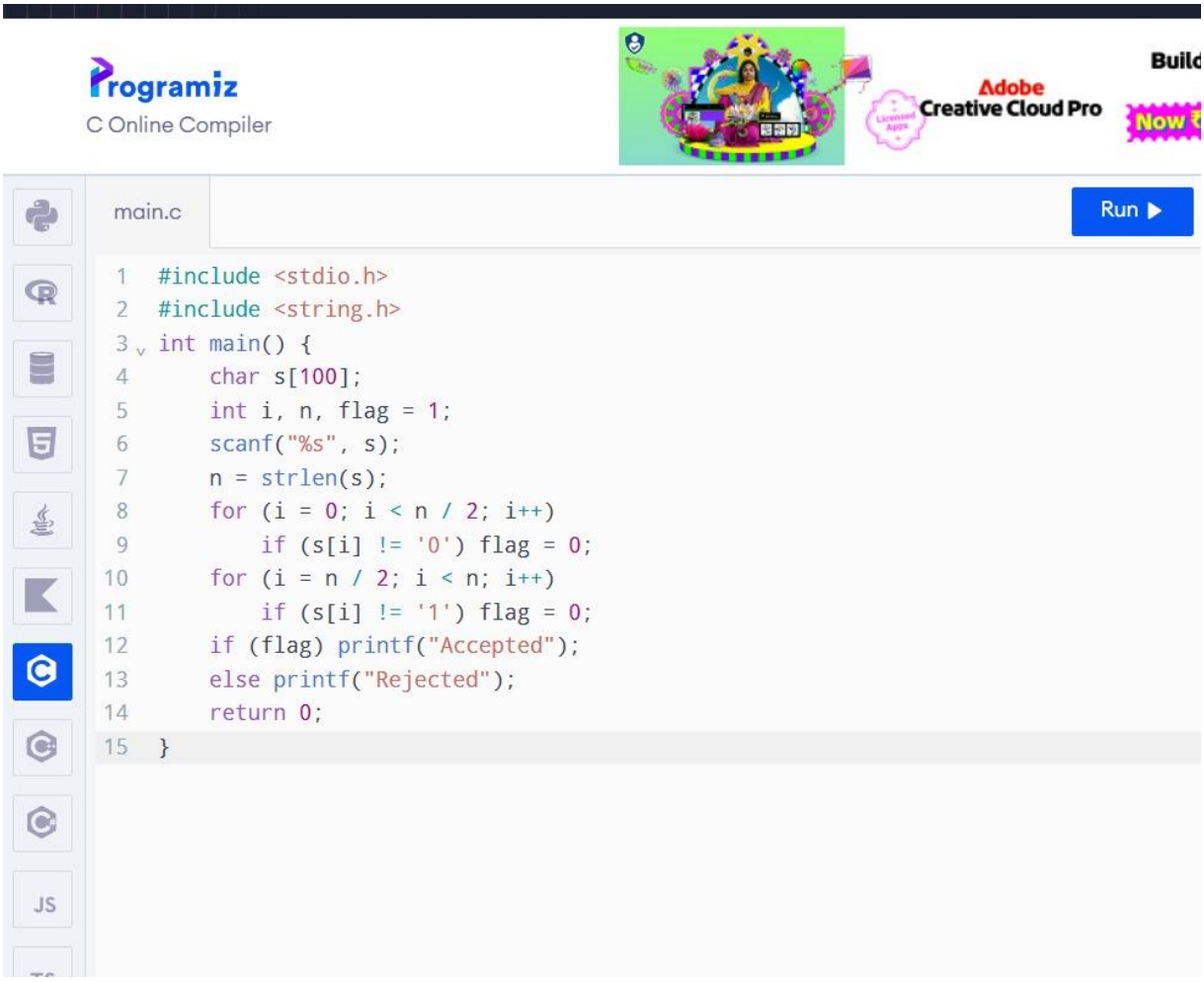
5. **Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG) S → 0S0 | A A → 1A | ε**

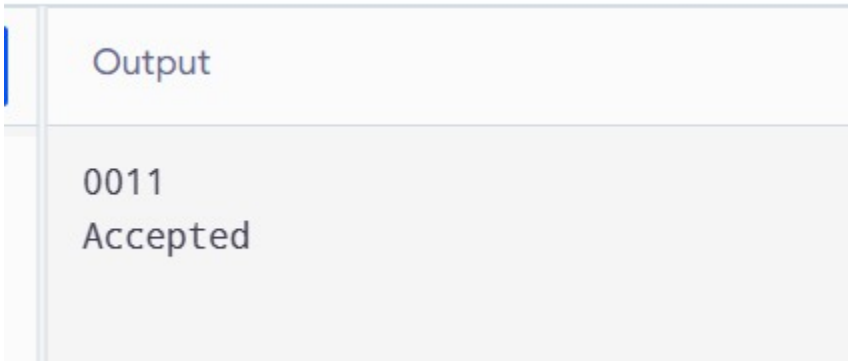
- Check whether the string has an even number of 0s surrounding a sequence of 1s. The zeros must appear in matching pairs at the beginning and end.

AIM: To write a C program to check whether a string belongs to the given CFG

CODE:



Output:



Result: The program accepts strings of the form \(0^n1^m0^n\), where \(n,m \geq 0\), and rejects other strings.